

TP1 - Semantic Network with Protege

1. Objective

The objective of this practical work (TP1) is to create a simple semantic network using the Protege tool. The chosen domain is a library system, and the ontology includes concepts such as Book, Author, User, and Borrowing. We aim to model relationships like who wrote a book and who borrowed it.

2. Ontology Structure

2.1 Classes

- Document
- Book (subclass of Document)
- Author
- User
- Borrowing

2.2 Object Properties

- hasAuthor (Book -> Author)
- isAuthorOf (inverse of hasAuthor)
- borrowedBy (Book -> User)
- hasBorrowed (inverse of borrowedBy)

2.3 Data Properties

- hasTitle (string)
- publicationYear (xsd:gYear)

2.4 Individuals

- TheLittlePrince (Book)
- SaintExupery (Author)
- PierreDurand (User)

-> Relations:

TheLittlePrince hasAuthor SaintExupery

TheLittlePrince borrowedBy PierreDurand

3. How to Visualize & Reason in Protege

1. Open Protege and load the OWL file.
2. Go to the 'OntoGraf' tab to view the concept map.
3. Enable the DL Query tab: Window > Tabs > DL Query.
4. Use queries like: Book, hasAuthor some Author, etc.

5. Go to Reasoner > Start reasoner (e.g., HermiT) to activate inference.

4. Logical Constraint

You can add a restriction on the class Book:

-> Book must have at least one author.

Create this restriction with:

hasAuthor min 1 Author

5. SPARQL Query Example

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

PREFIX owl: <http://www.w3.org/2002/07/owl#>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

PREFIX ex: <http://www.example.org/library#>

```
SELECT ?book ?author WHERE {  
  ?book a ex:Book .  
  ?book ex:hasAuthor ?author .  
}
```